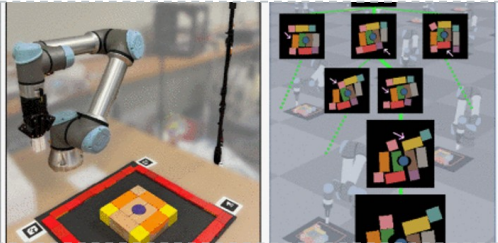
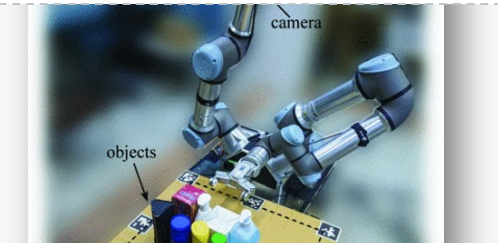
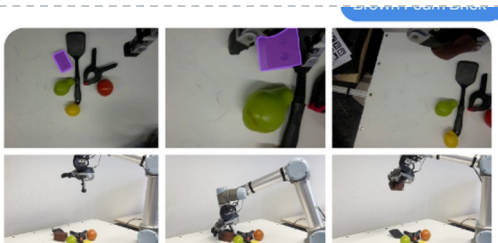


Related work

TRACE uses one privileged rollout to guide recurrent control and define a scene-specific execution horizon.

Method	Observation history	No sensing retractions required	No online search or simulation	Fixed rollout context	Rollout-defined execution horizon
object input					
 PMBS (previous SOTA) IROS 2022 · Complete	✗	✗	✗	✗	✗
 Kiatos et al. IROS 2024 · Partial	✗	✓	✓	✗	✗
 Mosbach & Behnke ICRA 2025 · Partial	✓	✓	✓	✗	✗
TRACE (Ours) Complete initially → Partial online	✓	✓	✓	✓	✓

Huang et al., PMBS: Parallel Monte Carlo Tree Search with Batched Rigid-Body Simulations for Speeding Up Long-Horizon Episodic Robot Planning, IROS 2022.